Name:_____ Period:_____
Earth and Space Science

MINERALS
LEARNING GUIDE

Learning Intention
Today we will be learning what makes a substance a mineral, how its atomic structure controls its properties, and how minerals form so we can identify minerals and explain how the crystals in places like Carlsbad Caverns and White Sands came to be.

Success Criteria
I can… ● Use the five parts of the definition of a mineral to decide whether a substance is a mineral. ● Explain how a mineral's atomic structure determines its properties. ● Describe how minerals form from cooling magma and from water, and use properties to identify a mineral.

Carlsbad Caverns: A World Built from Minerals

Carlsbad Caverns National Park, in southeastern New Mexico, protects more than 119 caves. Its Big Room is one of the largest cave rooms in North America. Hanging from the ceilings and rising from the floors are thousands of stone formations made of the mineral calcite. Each one grew slowly, one drop of water at a time. Not far away, at White Sands National Park, the dunes are made of another mineral, gypsum. In this guide, these two parks are our examples: they show what minerals are, how they form, and how we can tell them apart. As you read, keep this question in mind: *What makes something a mineral, and how can we tell one mineral from another?*

What Is a Mineral?

A **mineral** is a substance that meets all five of these requirements:

- **Naturally occurring:** It forms in nature, not in a factory or lab.
- **Inorganic:** It is not made by a living thing and is not made of the remains of living things.
- **Solid:** It has a definite shape and volume at Earth's surface.
- **Crystal structure:** Its atoms are arranged in an orderly, repeating pattern.
- **Definite chemical composition:** It is always made of the same elements in the same proportions. For example, calcite is always calcium carbonate ($CaCO_3$).

If a substance is missing even one of these, it is not a mineral. **Minerals are the building blocks of rocks.** A rock is made of one or more minerals. For example, granite is made mostly of the minerals quartz and feldspar.

Check for Understanding

1. Write **Y** if the substance is a mineral or **N** if it is not.

_____ a. Calcite crystals growing on a cave wall at Carlsbad

_____ b. Glass made in a factory

_____ c. Coal, formed from the remains of ancient plants

_____ d. Gypsum crystals at White Sands

_____ e. Granite, a rock made of several minerals

_____ 2. Obsidian is a volcanic glass that cools so quickly that its atoms do not form an orderly, repeating pattern. Obsidian is not a mineral because it does not have
(1) a natural origin
(2) a crystal structure
(3) a solid form
(4) any elements

Structure Determines Properties

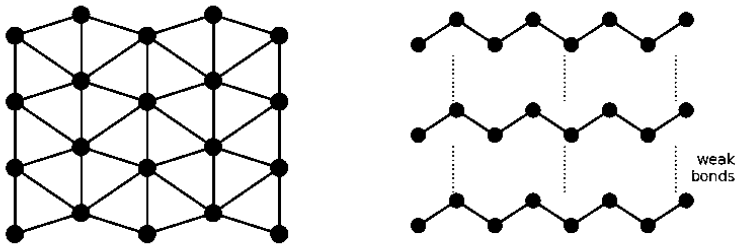
The way a mineral's atoms are arranged, and how strongly they are bonded, controls how the mineral looks and behaves. This includes its crystal shape, how hard it is, and how it breaks. For example, the atoms in halite (rock salt) are arranged in cubes, so halite forms cube-shaped crystals and breaks into cubes.

Diamond and graphite are both made of only one element, carbon. In **diamond**, every carbon atom is strongly bonded to its neighbors in all directions, making diamond the hardest natural mineral. In **graphite**, the carbon atoms form flat sheets. The bonds between the sheets are weak, so the sheets slide apart easily. That is why graphite is soft enough to use as pencil "lead."

Check for Understanding: Model It

The diagram represents the arrangement of carbon atoms in diamond and graphite. Black dots represent carbon atoms, and lines represent bonds.

(Not drawn to scale)



Diamond **Graphite**

_____ 1. Diamond and graphite are both made of carbon, but diamond is much harder. This difference is mainly caused by
(1) the color of each mineral
(2) the way the carbon atoms are arranged and bonded
(3) the amount of carbon in each mineral
(4) where each mineral is found

2. **Use the diagram** to explain why graphite is soft enough to use in pencils, while diamond is the hardest natural mineral.

_____

_____

Identifying Minerals by Their Properties

Because each mineral has its own structure and composition, each has its own set of properties. Geologists test several properties to identify a mineral.

Property	What it is	Example
Color	The color you see. Easy to observe, but often unreliable because small impurities change it.	Quartz can be clear, white, pink, or purple.
Streak	The color of a mineral's powder when rubbed on an unglazed tile (streak plate). More reliable than color. A streak plate has a hardness of about 7, so harder minerals leave no streak.	Hematite may look silver or black, but its streak is reddish-brown.
Luster	How the surface reflects light: **metallic** (shiny like metal) or **nonmetallic** (glassy, pearly, dull, and so on).	Pyrite has a metallic luster. Calcite has a glassy, nonmetallic luster.
Hardness	How well a mineral resists being scratched. A harder mineral scratches a softer one.	Gypsum can be scratched by a fingernail. Quartz scratches glass.
Cleavage	Breaking along smooth, flat surfaces where the bonds between atoms are weak.	Halite breaks into cubes.
Fracture	Breaking along rough or curved surfaces because the bonds are about equally strong in all directions.	Quartz breaks with curved, shell-like surfaces.
Crystal shape	The shape a crystal grows into. It is controlled by the arrangement of the mineral's atoms.	Halite grows as cubes. Quartz grows as six-sided columns with pointed ends.
Density	Mass ÷ volume. It stays the same no matter how big the sample is.	Pyrite (about 5.0 g/cm³) feels much heavier than a same-sized piece of quartz (about 2.7 g/cm³).
Other tests	Some minerals have special properties, such as reacting with acid or being magnetic.	Calcite bubbles (fizzes) in dilute acid. Magnetite attracts a magnet.

Hardness scale. Geologists compare hardness using the **Mohs Hardness Scale**, which is in your ESSRT. It ranks minerals from 1 (softest, talc) to 10 (hardest, diamond). It also lists common tools you can use to test hardness: a fingernail (2.5), a copper penny (3.5), a glass plate (5.5), and a steel nail (6.5). A mineral or tool can scratch anything with a lower hardness number.

Check for Understanding

1. Write the name of the **property** being tested in each example.

_____ a. A mineral leaves reddish-brown powder on a streak plate.

_____ b. A mineral scratches a piece of glass.

_____ c. A mineral splits into flat, smooth sheets.

_____ d. A mineral fizzes when a drop of acid is added.

_____ 2. Why is streak usually a better property than color for identifying a mineral?
(1) Streak is easier to see than color.
(2) The color of a mineral can change because of impurities.
(3) Every mineral has a white streak.
(4) Streak shows how hard a mineral is.

How Minerals Form

Minerals form in two main ways:

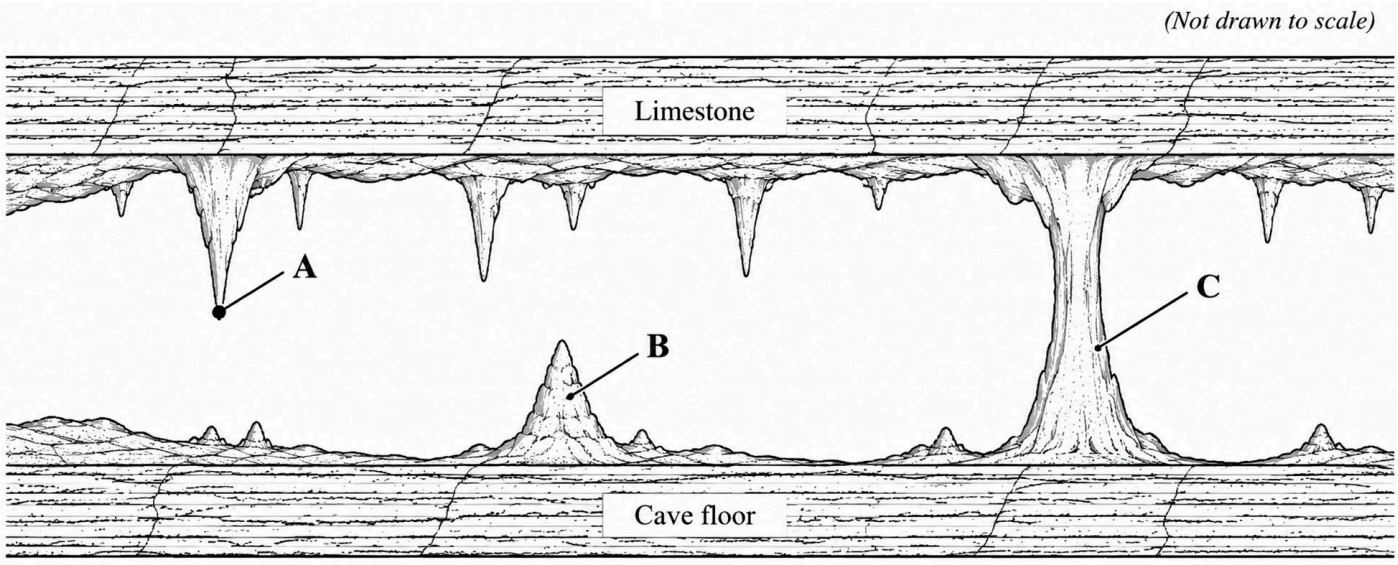
- **From magma or lava:** As melted rock cools, its atoms lock together into mineral crystals. You learned that this is called crystallization. Magma that cools slowly underground has time to grow larger crystals. You will learn more about this in the next learning guide.
- **From water:** Water can dissolve minerals. When the water evaporates or its conditions change, the dissolved material comes back out as solid crystals. This is called **precipitation**.

Carlsbad Caverns shows minerals forming from water. The caves were dissolved out of limestone by acidic water. Today, water that seeps through the limestone carries dissolved calcite. When a drop reaches the cave air, it loses some carbon dioxide gas and leaves behind a tiny ring of calcite. Drop by drop, a **stalactite** grows down from the ceiling. Where drops land on the floor, a **stalagmite** grows up. When a stalactite and stalagmite meet, they form a **column**.

White Sands shows minerals forming as water evaporates. Rain and melting snow dissolve gypsum from the nearby mountains and carry it into a basin with no outlet to the sea. When the water evaporates, gypsum crystals are left behind. Wind breaks the crystals into sand and piles it into dunes.

Check for Understanding: Model It

The diagram represents a cross section of part of a cave at Carlsbad Caverns.



1. **Label** each formation on the diagram as a **stalactite**, **stalagmite**, or **column**.

_____ 2. The gypsum crystals at White Sands formed mainly when
(1) magma cooled slowly underground
(2) lava cooled quickly at the surface
(3) water containing dissolved gypsum evaporated
(4) living things built shells

ESSRT Connection

Use the **Mohs Hardness Scale** in the 2026 Revised Edition Reference Tables for Earth and Space Sciences.

Mohs Hardness Scale

Hardness	Mineral Name	Tools
10.0	Diamond	
9.0	Corundum	
8.0	Topaz	
7.0	Quartz	
6.5		Steel nail
6.0	Orthoclase	
5.5		Glass plate
5.0	Apatite	
4.0	Fluorite	
3.5		Copper penny
3.0	Calcite	
2.5		Finger nail
2.0	Gypsum	
1.0	Talc	

_____ 1. A mineral scratches a glass plate but is scratched by a steel nail. Its hardness is between
(1) 2.5 and 3.5
(2) 3.5 and 5.5
(3) 5.5 and 6.5
(4) 6.5 and 10.0

_____ 2. Which tool would scratch calcite but not fluorite?
(1) finger nail
(2) copper penny
(3) glass plate
(4) steel nail

3. Gypsum and calcite are both white and both have cleavage. Name one tool from the scale that would scratch gypsum but not calcite.

_____

Cluster Set 1

Base your answers to questions 1 through 3 on the passage and data table below and on your knowledge of Earth and Space Science.

Herkimer "Diamonds" vs. True Diamonds

In Herkimer County, New York, people dig for sparkling, clear crystals known as Herkimer "diamonds." The crystals form in small openings in rock and often have points on both ends. At first glance, they look like real diamonds. However, Herkimer "diamonds" are actually a variety of the mineral quartz. The table compares the two minerals.

Property	Herkimer "diamond" (quartz)	True diamond
Appearance	Clear, glassy crystal	Clear, glassy crystal
Chemical composition	Silicon and oxygen (SiO_2)	Carbon (C)
Hardness (Mohs)	7	10
Breakage	Fracture (curved surfaces)	Cleavage (flat surfaces)

_____ 1. Which claim is best supported by the passage and table?
(1) Herkimer "diamonds" and true diamonds are the same mineral.
(2) Minerals that look alike can have different compositions and properties.
(3) Hardness is determined by a mineral's color.
(4) All clear minerals are made of carbon.

2. Identify **two properties** from the table that can be used to tell a Herkimer "diamond" from a true diamond, and describe how each property is different.

Property 1: _____ How it differs: _____

Property 2: _____ How it differs: _____

3. A student finds a clear, glassy crystal. It scratches a steel nail and breaks along curved surfaces. Identify the mineral and give **two pieces of evidence** from the table.

Mineral: _____

Evidence 1: _____

Evidence 2: _____

Mini Glossary

Cleavage – the tendency of a mineral to break along smooth, flat surfaces

Column – a cave formation made when a stalactite and a stalagmite grow together

Crystal shape – the shape a mineral crystal grows into, controlled by its atomic arrangement

Crystal structure – an orderly, repeating arrangement of atoms

Crystallization – the formation of mineral crystals as magma or lava cools

Definite chemical composition – always made of the same elements in the same proportions

Fracture – the tendency of a mineral to break along rough or curved surfaces

Hardness – a mineral's resistance to being scratched

Inorganic – not made by a living thing or from the remains of living things

Luster – the way a mineral's surface reflects light (metallic or nonmetallic)

Mineral – a naturally occurring, inorganic solid with a crystal structure and a definite chemical composition

Mohs Hardness Scale – an ESSRT scale from 1 to 10 that compares the hardness of minerals and common tools

Precipitation (of minerals) – the forming of solid crystals from minerals that were dissolved in water

Stalactite – a cave formation that hangs down from the ceiling

Stalagmite – a cave formation that grows up from the floor

Streak – the color of a mineral's powder when rubbed on a streak plate

Streak plate – an unglazed porcelain tile, hardness about 7, used to test streak